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A transformer-based intelligent online virtual conductor assistance system for choirs

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Abstract

The development of virtual technology and artificial intelligence has made the automation and intelligence of choral conducting an important direction for enhancing the expressiveness of online choral performances. Existing methods, however, often fail to generate conducting gestures that are both natural and closely synchronized with music. To this end, this study proposes a Conformer-based Virtual Conductor Motion Generation Model (CV-CMGM) based on the Convolution-Enhanced Transformer (Conformer), which is used to automatically generate a sequence of 3D conductor motions from music signals. The CV-CMGM model consists of a music and motion encoder, a cross-modal generator, and a local temporal and global content discriminator. The Conformer structure captures local and global temporal dependencies in music and motion sequences, enabling cross-modal generation from music to conductor motions. Experimental results on the ConductorMotion100 dataset show that CV-CMGM achieves a motion Fréchet Inception Distance (FID) of 27.96, a geometric FID of 19.35, motion diversity and geometric diversity of 5.94 and 5.12, respectively, and a beat alignment score of 0.214. CV-CMGM outperforms the comparative models in all indicators. User studies further indicate that the virtual conductor motions generated by CV-CMGM achieve scores above 4.4 in interpretability, dynamic expressiveness, musical structure coherence, and naturalness/smoothness, outperforming other models. These data indicate that the proposed method can effectively generate virtual conducting motions that match the musical content and exhibit good expressiveness.

Keywords Choral conducting, Virtual conducting, Transformer, Motion generation, Music-motion alignment

1 Introduction

With the rapid advancement of artificial intelligence and deep learning technologies, virtual conducting has gained increasing attention in the domains of music education and performance. Choral conducting—an integral aspect of musical interpretation—requires a high degree of musical comprehension, coordination, and physical expressiveness from the conductor [1, 2]. Traditional choral performances depend on the conductor's real-time gestures to shape ensemble phrasing and emotional expression. Fueled by ongoing developments in information technology, particularly the emergence of intelligent



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online instruction, virtual choral conducting has become a prominent area of research. Through virtual platforms, choirs can now receive real-time, precise conducting cues, facilitating remote music instruction and enabling intelligent conducting assistance.

In recent years, VR and deep learning technologies have been widely applied in music education, dance generation, and motion recognition. Sun [3] combined the advantages of deep learning neural networks to analyze the impact of VR music education on learning outcomes. They expected that VR would provide an immersive experience that could significantly improve students' music proficiency, music theory understanding ability, and overall learning participation. Yan and Xia [4] applied VR and artificial intelligence technologies to the music teaching process, constructed an interactive course teaching mode, and verified the feasibility of combining VR and AI technologies with music teaching through experimental analysis. In the domain of dance generation, Jiang and Yan [5] proposed a dance consistency motion generation model that created movement patterns through latent space interpolation and adjusted motion amplitude to enhance realism. Kim and Lee [6] introduced an autoregressive K-pop dance generation model, which improved the ability to generate synchronized dance sequences through additional learning strategies. Regarding choral conducting motion, Zhang [7] developed a hybrid convolutional neural network (CNN) model that integrated spatial and temporal information to perform pose recognition, thereby enhancing the real-time performance and accuracy of choir gestures.

Although these studies have achieved progress in virtual motion generation and pose recognition, limitations remain. Most methods focus on dance or unimodal motions and lack cross-modal modeling specifically for choral conducting gestures. Furthermore, the naturalness and musical alignment of the generated conducting motions still require improvement [8–10]. Transformer-based motion-generation frameworks have recently been applied to virtual conducting in an effort to produce gestures that feel both natural and musically grounded [11, 12]. Building on the architecture's success in natural-language processing, researchers have adapted Transformers to a variety of sequence-modelling tasks, including music generation. The Conformer- a convolution-augmented Transformer- further improves performance by capturing long-range temporal structure while preserving fine-grained local detail [13, 14]. Despite these advances, synchronizing musical cues with conductor-gesture sequences remain challenging. The model must fuse heterogeneous information streams (audio features, score annotations, beat positions, and motion trajectories) and learn their complex, context-dependent alignments. Designing architectures that reconcile these demands- without sacrificing gesture naturalness or rhythmic precision- is still an open problem in virtual-conducting research.

In real-world choir rehearsals and performances, the conductor guides the choir's rhythm and emotional expression through gestures, effectively "conducting the music" [15]. In contrast, in online virtual choir conducting assistance scenarios, it is necessary to perceive and analyze the musical signal in real time or near real time, and to generate corresponding conducting motions based on the current rhythmic structure and dynamic changes of the music. This enables guidance and support for virtual choir performances or remote teaching sessions. Motivated by these practical requirements, this study adopts a "music-to-conductor-motion" modeling approach, simulating the immediate gestural responses that human conductors produce in response to perceived musical structure and performance status. This approach does not diminish the conductor's

leading role in choir performance; rather, it addresses the needs of online virtual systems and real-time interaction. It provides a feasible, interactive, and expressive method for generating intelligent conducting motions in virtual choir environments, thereby enhancing both the guidance capability and the user experience of virtual choir systems. This study introduces an autoregressive, generative-adversarial framework built on the Conformer architecture to generate virtual conducting-gesture sequences that remain tightly synchronized with the music. The generator contains three key components:

- (1) Dual encoders that separately embed musical features and conducting motions;
- (2) A cross-modal fusion module that merges these representations; and
- (3) A gesture-generation sub-network that outputs the final motion sequence.

A dual-scope discriminator then evaluates the results, performing both local temporal checks (to enforce beat-level accuracy) and global content checks (to preserve expressive coherence). This design provides a novel roadmap for virtual-conducting systems and offers a solid theoretical and practical foundation for intelligent support in online music education and virtual choir performance. The experimental results demonstrate that the proposed method can generate natural, coherent, and expressive conducting gestures, outperforming existing baseline methods in terms of gesture authenticity, gesture diversity, and music-gesture synchronization. This study provides an effective technical solution for virtual choral conducting systems and shows promising potential in application scenarios such as online music education, virtual choral performances, and intelligent music interaction systems. The study also offers new research ideas and implementation paths for the development of future virtual conducting technologies.

The primary contributions of this study are summarized as follows:

First, an autoregressive generative adversarial network based on the Conformer architecture is proposed, capable of jointly modeling musical signals and conductor motion sequences. Compared with traditional approaches, this method exhibits superior performance in cross-modal representation and temporal consistency.

Second, a Conformer-based music and motion encoder, a cross-modal generator, and local temporal and global content discriminators are designed. By incorporating historical motion priors and cross-modal attention mechanisms, the generated conductor motions achieve higher rhythmic responsiveness, greater motion diversity, and improved naturalness compared with existing methods.

Finally, systematic experiments conducted on the ConductorMotion100 dataset demonstrate that the model significantly outperforms baseline methods in motion realism, motion diversity, and music–motion synchronization. These results highlight the model's practical applicability and potential for virtual choir conducting assistance, providing theoretical guidance and implementation reference for the development of online virtual conducting systems.

2 Materials and methods

2.1 Theoretical foundations of the convolution-enhanced transformer

The Transformer and CNN models offer complementary strengths. Transformers excel at capturing long-range global dependencies, while CNN effectively extracts local features through their receptive fields. To leverage the advantages of both architectures, Gulati et al. [16] proposed the Conformer, which integrated convolutional modules into

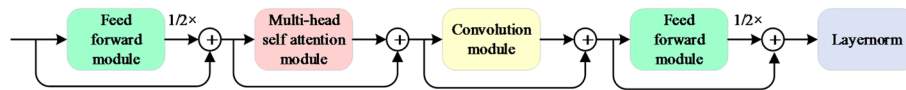


Fig. 1 Illustration of the conformer architecture

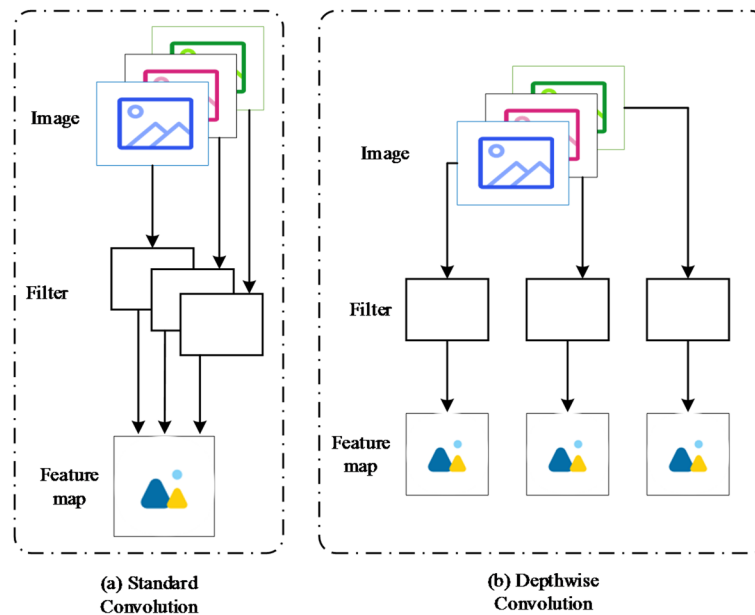


Fig. 2 Illustration of depthwise convolution **a** traditional convolution operation; **b** depthwise convolution operation

the Transformer structure to enhance the modeling of local feature patterns. Figure 1 illustrates the overall Conformer architecture.

In the Conformer architecture, the convolutional module differs from traditional CNN by employing lightweight depthwise and pointwise convolutions. Depthwise convolution applies separate convolution operations to each channel independently, reducing redundant computations and parameter count. Pointwise convolution uses 1×1 kernels to integrate information across channels, enabling effective feature fusion. This convolutional design allows the model to capture local patterns in the input sequence more efficiently while preserving the ability to model global dependencies [17, 18]. The structure of Depthwise Convolution is illustrated in Fig. 2.

Pointwise Convolution, similar to traditional convolution, uses a 1×1 kernel size. Its purpose is to process the channels of each pixel point without involving the computation of neighboring pixels, which helps preserve local information while increasing the dimensionality of the features [19, 20]. By combining CNN and Transformer, the Conformer enhances the ability to process local information while strengthening the modeling capacity for global temporal features.

For this study, the Conformer architecture offers an ideal network for multimodal temporal modeling, capable of simultaneously processing music signals and conductor motion sequences, both of which exhibit strong temporal dependencies. In the virtual choir conducting scenario, music signals serve as known inputs, representing beat structures, dynamic variations, and overall emotional contours, while conductor motion sequences reflect human expressive patterns under given musical conditions. The

convolutional module in Conformer captures local temporal variations, such as motion adjustments around beat points, whereas the self-attention mechanism models long-range dependencies, including phrase-level conducting structures. This design enables the extraction of key musical information, such as rhythm and dynamic changes, while learning continuity and postural variation patterns from historical motion sequences. By integrating local feature modeling with global temporal dependencies, Conformer facilitates the generation of virtual conductor motions that are temporally coherent, expressively natural, and semantically consistent with conducting gestures, providing a solid foundation for subsequent virtual choir conducting motion generation.

2.2 Conformer-based virtual conducting motion generation model

To achieve high-quality virtual conducting motion generation, a Conformer-based Virtual Conducting Motion Generation Model (CV-CMGM) is proposed. This model consists of three main modules: the music and motion encoder, the cross-modal generator, and the local temporal and global content discriminator, as illustrated in Fig. 3.

In this study, virtual conductor motions are represented as sequences of 3D skeletal joint positions. Specifically, each conducting motion is modeled as a 3D skeleton consisting of J key joints, with the state of each frame represented by the 3D coordinates of all joints in a unified reference frame. For a motion sequence of length T , the sequence is denoted as $C = \{c_1, c_2, \dots, c_T\}$, where $c_t \in \mathbb{R}^{J \times 3}$. The musical input is temporally aligned with the motion sequence. Music signals are first segmented into frames and processed at the same frame rate as the motion sequence, ensuring a one-to-one correspondence between musical features and conducting motions in the temporal dimension. For all experiments, both music and motion sequences are handled in fixed-length segments of length T during training and generation. The model takes as input the sequence of music features along with the historical conductor motion sequence and outputs the predicted 3D skeletal sequence for future time steps.

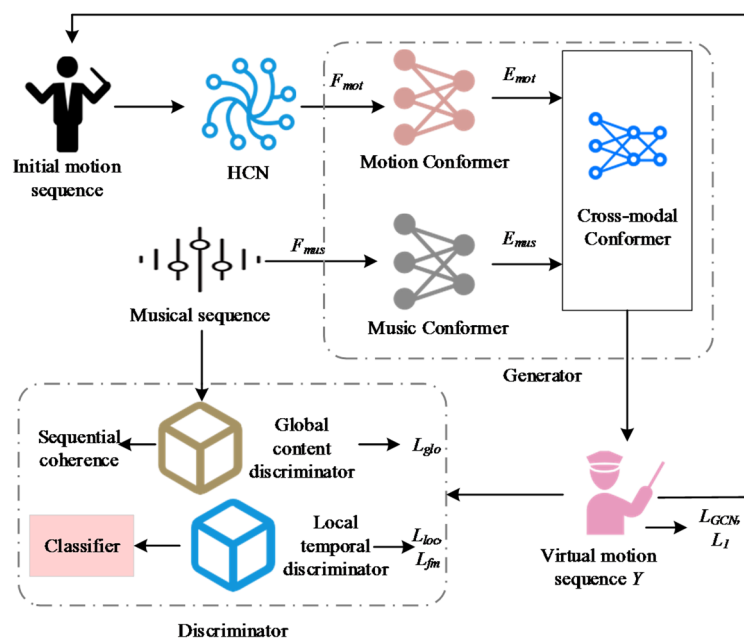


Fig. 3 Music and virtual conducting motion generation model

(1) Music and Motion Encoder.

To effectively extract features from both music signals and conducting motion sequences, the CV-CMGM model incorporates two distinct encoders, each designed to process music and motion features separately.

For music features, a music Conformer is used as the encoder, capturing both local spatial and global temporal characteristics of the music sequence. This encoder consists of a multi-head attention mechanism, convolutional layers, and feed-forward layers. Musical features serve not only as the driving signal for motion generation but also as a representation of the rhythmic structure, dynamic variations, and overall emotional contour of the music, providing both temporal constraints and expressive references for generating conductor motions. Given a music sequence of length T , denoted as $M = \{m_1, m_2, \dots, m_T\} \in \mathbb{R}^{T \times D}$, where D represents the dimensionality of the music features per frame, Mel-Frequency Cepstral Coefficients (MFCCs) are first extracted to obtain the feature sequence $MFCC = \{MFCC_1, MFCC_2, \dots, MFCC_T\} \in \mathbb{R}^{T \times D}$. To enhance the model's perception of temporal rhythm, absolute positional encodings (PE) are added to the input prior to the multi-head attention mechanism, resulting in the position-aware music feature representation F_{mus} , as shown in Eq. (1).

$$F_{mus} = MFCC + PE \quad (1)$$

Subsequently, the music features are further processed using a combination of multi-head attention mechanisms, convolutional layers, and feed-forward layers to obtain the final embedding vector E_{mus} , which captures both the global temporal characteristics and local information of the music. The processing steps can be represented as Eqs. (2)–(4):

$$F'_{mus} = F_{mus} + ATT(F_{mus}) \quad (2)$$

$$F''_{mus} = F'_{mus} + Conv(F'_{mus}) \quad (3)$$

$$E_{mus} = F''_{mus} + FFN(F''_{mus}) \quad (4)$$

Here, $ATT(\cdot)$, $Conv(\cdot)$, and $FFN(\cdot)$ denote the multi-head attention mechanism, convolutional layer, and feed-forward neural network operations, respectively.

Similarly, a motion Conformer is designed as a feature encoder for the conducting motion sequence. This encoder employs a combination of masked multi-head attention mechanisms and convolutional layers, while simultaneously incorporating both absolute and relative position encodings to enhance the model's ability to capture the temporal characteristics of the motion sequence. Conductor motion sequences are represented as 3D skeletal joint positions, denoted as $C = \{c_1, c_2, \dots, c_T\}$, $c_t \in \mathbb{R}^{J \times 3}$, where J is the number of key joints and each frame c_t encodes the 3D coordinates of all joints in a world reference frame. Local motion features are first extracted from the sequence using a Hierarchical Convolutional Network (HCN), yielding the processed motion features $\tilde{C} = \{\tilde{c}_1, \tilde{c}_2, \dots, \tilde{c}_T\}$, which serve as the input features $F_{mot} = \tilde{C}$ for the motion Conformer encoder. To ensure the coherence and plausibility of generated motions, the motion encoder incorporates motion priors learned from historical sequences, constraining the outputs in terms of continuity, joint amplitude, and conductor-specific semantic consistency. The motion features are processed through a combination of

masked multi-head attention and convolution layers, while absolute and relative positional encodings are introduced to capture both local variations and long-range dependencies. The final motion embedding vector is expressed as Eq. (5):

$$E_{mot} = F_{mot} + ATT(F_{mot}) + Conv(F_{mot} + ATT(F_{mot})) \quad (5)$$

(2) Cross-modal generator

In virtual choir conducting scenarios, there is no strict one-to-one correspondence between music and conductor motions [21]. To address this, a cross-modal Conformer generator is designed to produce motion sequences that align with the statistical distribution of conductor actions while respecting the temporal structure of the music. This module does not simply map music features to motions; instead, it incorporates historical motion sequences as references, ensuring that generated actions maintain natural continuity, semantic plausibility, and consistency with the underlying musical dynamics. The music embedding vector E_{mus} is used as the query (Q) in the multi-head attention mechanism, while the historical motion embedding vector E_{mot} serves as the key (K) and value (V), effectively capturing the cross-modal interaction between the two. Its structure is illustrated in Fig. 4.

After processing through the multi-head attention mechanism, the features are further processed by convolutional layers and feed-forward layers, resulting in an intermediate embedding vector E_{cross} . To enhance the model's expressive capability, the cross-modal Conformer introduces a fully connected layer and generates the final motion sequence $Y = \{y_1, y_2, \dots, y_T\} \in \mathbb{R}^{T \times D}$ through the tanh activation function. The generation process is represented as Eqs. (6)–(9):

$$E_{cross} = E_{mot} + ATT(E_{mus}, E_{mot}) \quad (6)$$

$$E'_{cross} = E_{cross} + Conv(E_{cross}) \quad (7)$$

$$E''_{cross} = E'_{cross} + FFN(E'_{cross}) \quad (8)$$

$$Y = \tanh(LN(E''_{cross})) \quad (9)$$

In this context, $LN(\cdot)$ denotes layer normalization. Ultimately, through the tanh function and the fully connected layer, the future conducting motion sequence Y is generated, which aligns with the input music and historical conducting movements, ensuring

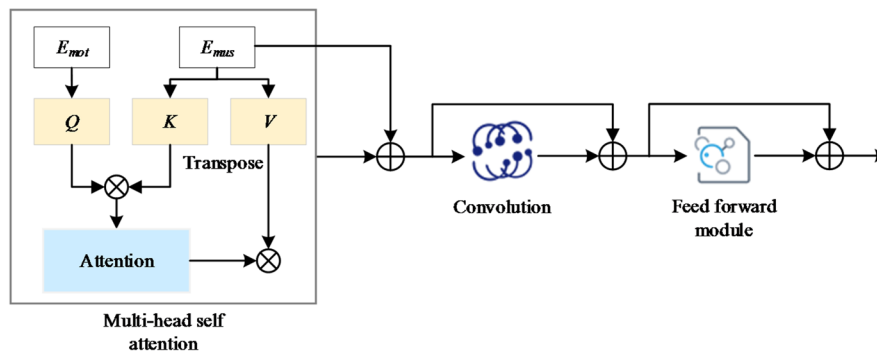


Fig. 4 Cross-modal conformer network architecture

that the generated virtual conducting movements adhere to the rhythmic, stylistic, and temporal requirements of the choral performance.

(3) Local temporal and global content discriminators

To ensure that the generated virtual conducting motion sequence is temporally coherent and consistent with the musical rhythm, the CV-CMGM model incorporates two key discriminators: the local temporal discriminator and the global content discriminator.

The local temporal discriminator aims to assess the temporal coherence of the generated virtual conducting motion sequence. Specifically, when a conducting issues a particular instruction, the hand movements should exhibit a certain temporal pattern and sequence. To achieve this, the generated virtual conducting motion sequence $Y \in \mathbb{R}^{T \times D}$ is divided into K overlapping subsequences. Each subsequence is then input to the motion discriminator for processing, producing a score that indicates whether the subsequence meets the expected temporal correlation. Ultimately, the local temporal discriminator evaluates the overall temporal coherence of the entire motion sequence based on the K scores, ensuring that the conducting motion at each moment is consistent with the preceding and succeeding motions.

The global content discriminator ensures the global consistency between the generated virtual conducting motion sequence and the musical rhythm and style. The input to this discriminator consists of the music features $M \in \mathbb{R}^{T \times D}$ and the generated virtual conducting motion sequence $Y \in \mathbb{R}^{T \times D}$. For the conducting motion sequence Y , it is first encoded by the motion discriminator to obtain a motion feature vector $E_{mot} \in \mathbb{R}^{256}$. For the music signal M , a one-dimensional convolutional network is used for initial processing, followed by a bidirectional gated recurrent unit (GRU) network to extract the temporal features of the music, resulting in a music feature vector $E_{mus} \in \mathbb{R}^{T \times 256}$. Subsequently, a self-attention mechanism is applied to further process the music features E_{mus} , producing the final music feature vector $E'_{mus} \in \mathbb{R}^{256}$.

The generated motion feature vector E_{mot} and the processed music feature vector E'_{mus} are concatenated and input into a classifier, which determines whether the generated virtual conducting motion sequence Y aligns with the global structure of the music.

(4) Loss function

To optimize the Conformer-based virtual conducting motion generation model, a combination of perceptual loss, GAN loss, L1 loss, and feature-matching loss is employed. This design ensures that the generated virtual conducting motion sequences maintain temporal coherence while accurately reflecting the rhythm and expressive style of the music.

The perceptual loss is computed by extracting high-level motion features from both the generated and ground-truth motions using a pre-defined Graph Convolutional Network (GCN). This loss measures the semantic difference between the generated and real motions in the action space, and is defined as Eq. (10):

$$L_{GCN} = \sum_{i=1}^n \lambda \|\varphi(P_i) - \varphi(G(M_i, X_i))\|_2 \quad (10)$$

In Eq. (10), P_i is the real conducting motion at the i -th frame, $G(M_i, X_i)$ is the virtual conducting motion generated by the model, φ is the feature extractor implemented via

the GCN, and λ is a hyperparameter used to balance the contribution of each layer to the loss. n is the number of samples.

To make the generated virtual conducting motion sequence more realistic, GAN training is applied, with the generator continuously improving based on feedback from the discriminator. The GAN loss includes the local temporal discriminator loss L_{loc} and the global content discriminator loss L_{glo} , as shown in Eq. (11).

$$L_{GAN} = L_{loc} + L_{glo} \quad (11)$$

The local temporal discriminator loss L_{loc} is defined as Eq. (12):

$$L_{loc} = E_P [\log D_1 (S (P))] + E_{X,M} [\log(1 - D_1 (S (X)))] \quad (12)$$

The global content discriminator loss L_{glo} is given by Eq. (13):

$$L_{glo} = E_{P,M} [\log D_2 (S (P, M))] + E_{X,M} [\log(1 - D_2 (S (X, M)))] \quad (13)$$

Here, D_1 and D_2 represent the local temporal discriminator and the global content discriminator, respectively, and $S(\cdot)$ is the function that samples the entire conducting motion sequence using a sliding window approach.

The L_1 loss is introduced to measure the difference between the generated and real motions, ensuring that the generated virtual conducting motion sequence closely resembles the real conducting motion sequence. Specifically, for the generated virtual conducting motion sequence and the real conducting motion sequence Y' , the L_1 loss is defined as Eq. (14):

$$L_1 = \sum_{i \in [0, 2]} \|Y'_i - Y_i\|_1 \quad (14)$$

In Eq. (14), Y'_i and Y_i represent the real and generated virtual conducting motions at the i -th frame, respectively.

Feature matching loss is employed to ensure that the generated virtual motion sequence is consistent with the features of the music at each layer of the network output [22, 23]. This can be expressed as Eq. (15):

$$L_{fm} = \sum_{k=1}^n \|D_{glo,k} (P, M) - D_{glo,k} (G (M), M)\|_2 \quad (15)$$

In Eq. (15), n represents the number of layers in the global content discriminator D_{glo} , $D_{glo,k}$ is the output of the k -th layer of the discriminator, and $G (M)$ denotes the generated choral conducting motion sequence.

The final loss function is the weighted sum of the individual loss terms, expressed as Eq. (16):

$$L = w_1 L_{GCN} + w_2 L_{GAN} + w_3 L_1 + w_4 L_{fm} \quad (16)$$

In Eq. (16), w_1 , w_2 , w_3 , and w_4 are the hyperparameters representing the weights for each loss component.

2.3 Virtual conducting assistance mode based on the virtual conducting motion generation model

Building upon the aforementioned CV-CMGM model, a virtual conducting assistance mode is proposed. This mode is capable of automatically generating conducting motions that align with the music's rhythm and style through a computer, thereby supporting traditional conducting tasks. In this mode, the input music and historical conducting motions are processed by the CV-CMGM to generate virtual conducting motions that match the music. Specifically, music features are encoded into temporal features by the music Conformer, while historical conducting motions are used by the motion Conformer to extract spatial and temporal features. The cross-modal Conformer combines the embedded representations of both, generating a conducting motion sequence corresponding to the current musical segment. The generated virtual conducting motion sequence naturally reflects changes in the music's rhythm, and with the aid of the local temporal and global content discriminators, it ensures the coherence and rationality of the motion sequence.

This virtual conducting assistance mode provides real-time, accurate conducting motion feedback to choir members with real-time, accurate conducting motion feedback. It also automatically adjusts the speed, intensity, and amplitude of conducting gestures based on the emotional expression of the music. For example, in a choral performance, as the rhythm of the music accelerates, the generated virtual conducting motions will automatically adjust to faster gestures to ensure synchronization among the choir members. Through this intelligent conducting assistance, the conducting can effectively reduce the repetitive aspects of conducting work, enhancing the choir's performance quality and overall collaboration efficiency.

2.4 Data collection and experimental setup

To evaluate the performance of the proposed CV-CMGM model, publicly available AIST++ (A Large-Scale Dance Motion Dataset with Music for Deep Learning) and ConductorMotion100 datasets are used for training and testing. The AIST++ dataset is a large-scale 3D human dance motion dataset that contains diverse 3D dance sequences accompanied by corresponding music segments, providing rich motion patterns and aligned musical information for the experiment [24]. Although AIST++ primarily focuses on dance motion, it offers strong representativeness for modeling temporal continuity, rhythm perception, and music-motion alignment. Consequently, the CV-CMGM model is pretrained on AIST++ to learn general temporal relationships between music and human motion, as well as motion generation priors.

For model evaluation, the ConductorMotion100 dataset is employed. This publicly available dataset contains 100 conducting motion sequences performed by professional conductors, with each sequence strictly aligned to the corresponding musical segment. It emphasizes motions of the upper limbs, arms, and torso, which are closely related to expressive conducting gestures. The dataset accurately captures dynamic characteristics of conducting, including beat indication, expressive intensity, and structural guidance. Motions are represented as 3D skeletal joint sequences, making the dataset suitable for deep learning-based conducting motion analysis and generation. For evaluation, ConductorMotion100 is split into training, validation, and test sets in a 70–15–15% ratio,

allowing systematic assessment of the model's generation quality and music-motion alignment in virtual conducting scenarios.

During data preprocessing, both the AIST++ and ConductorMotion100 datasets are mapped to a consistent 3D skeletal representation and processed within the same world coordinate system at a fixed frame rate. All motion sequences are normalized prior to input into the model to eliminate the influence of individual body shape differences on training and evaluation. Music signals are segmented to match the frame rate of the motion sequences, and MFCCs are extracted and augmented with absolute positional encodings to ensure strict temporal alignment between musical features and conducting motions. The model is trained in two stages: pretraining on AIST++ and fine-tuning on ConductorMotion100. Input sequences are generated using a fixed-length sliding window, and mini-batch training with a batch size of 32 is employed. The CV-CMGM model is trained end-to-end, without requiring separate pretraining for the generator or discriminator. The generator loss combines discriminator feedback with reconstruction error, incorporating local temporal loss and global content loss to enforce naturalness and music-motion consistency. The discriminator loss evaluates both local motion segments and global sequences to ensure temporal continuity and overall expressive coherence. During training, the generator and discriminator are optimized using Adam with learning rates of 0.0004 and 0.0001, respectively. Weight decay and gradient clipping are applied to prevent overfitting and gradient explosion. Training proceeds for 200 epochs, with each epoch representing a complete pass through the training set. Performance on the validation set is monitored using motion Fréchet Inception Distance (FID) and geometric FID to select the best model. Furthermore, motion and music sequences maintain strict temporal alignment within each batch, and all input and output tensors explicitly specify key dimensions, including frame length T , number of joints J , and feature dimension D . The parameter settings for the CV-CMGM model are shown in Table 1.

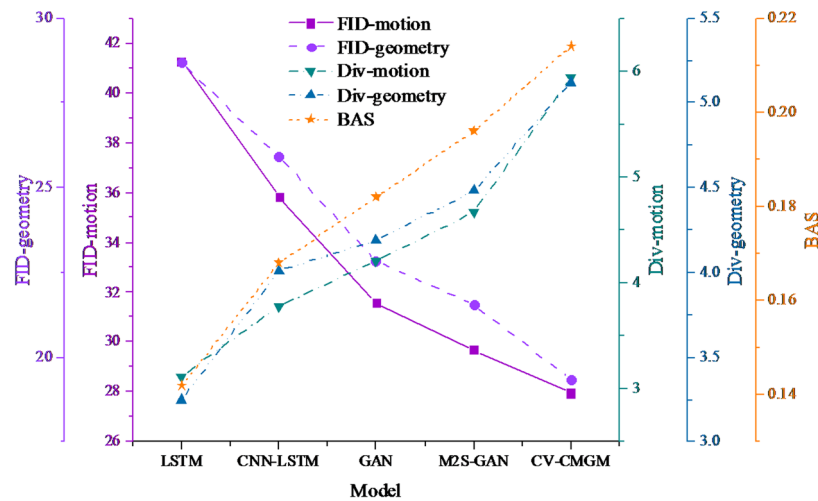
In the experiments, the effectiveness of the proposed CV-CMGM model for virtual choir conducting motion generation is evaluated from three perspectives: motion quality, motion diversity, and music–motion synchronization. Several baseline methods are selected for comparison, including Long Short-Term Memory (LSTM) networks, CNN-LSTM, GAN, and the Music Motion Synchronized Generative Adversarial Network (M2S-GAN) [25]. To ensure fair comparison, all baselines are retrained under the same data splits, sequence lengths, musical input conditions, and evaluation protocols. Each method uses identical music feature representations and skeletal motion definitions, and all are fine-tuned on the ConductorMotion100 dataset to avoid evaluation bias caused

Table 1 Model parameter settings

Model parameter	Setting value
Number of conformer network layers	5
Dimension of multi-head attention Q, K, V vectors	64
Number of attention heads	8
Feedforward layer structure	Two fully connected layers, GELU activation function, dropout layer
Generator learning rate	0.0004
Discriminator learning rate	0.0001
Optimizer	Adam

Table 2 Quantitative evaluation of generated virtual conducting motions

Model	FID-Motion	FID-Geom	Div-Motion	Div-Geom	BAS
LSTM	41.23	28.67	3.12	3.25	0.142
CNN-LSTM	35.84	25.91	3.78	4.01	0.168
GAN	31.57	22.83	4.21	4.19	0.182
M ² S-GAN	29.68	21.54	4.67	4.48	0.196
CV-CMGM	27.96	19.35	5.94	5.12	0.214

**Fig. 5** Performance comparison of models on conductormotion100

by domain discrepancies. Evaluation metrics include FID, Diversity (Div), and BAS. Both FID and Div are assessed from motion and geometric perspectives. FID is computed based on motion sequence embeddings: a pretrained motion recognition network (trained on AIST++) extracts high-level features from both generated and real motions, and the Fréchet distance between their distributions is calculated to quantify the realism and consistency of the generated motions. Geometric FID evaluates the spatial plausibility of generated motions based on the 3D joint position sequences. The Div metric measures the model's ability to generate diverse conducting motions while maintaining musical consistency by calculating the average pairwise distance in feature space. To further assess music–motion synchronization, BAS is introduced. This metric quantifies temporal alignment by detecting correspondences between musical beat positions and key velocity changes or rhythmic movements in the generated conducting sequences, providing an objective evaluation of music–motion temporal coherence. Additionally, a user experiment was conducted, inviting 10 professional conductors to rate the generated virtual conducting motions based on criteria such as comprehensibility, dynamic expressiveness, alignment with the musical structure, and natural fluency. Ratings were given on a 1 to 5 scale to validate the practical utility of the generated motions.

3 Results

3.1 Model performance evaluation

On the ConductorMotion100 dataset, the quantitative evaluation results of the generated virtual conducting motions are presented in Table 2; Fig. 5.

The results in Table 2; Fig. 5 reveal clear differences in the performance of the models. In terms of motion realism, CV-CMGM achieves the lowest values for both motion

FID (27.96) and geometric FID (19.35), indicating that the generated motions are closest to real conducting motions in terms of overall motion distribution and spatial structure. Traditional LSTM and CNN-LSTM models perform the worst, suggesting that purely sequential modeling or local convolutional features are insufficient to capture the complex temporal and spatial patterns of conducting movements. Regarding motion diversity, CV-CMGM attains the highest Div-Motion (5.94) and Div-Geom (5.12), outperforming M2S-GAN by 27.2% and 14.3%, respectively. This demonstrates that cross-modal modeling and the incorporation of historical motion priors effectively enhance the diversity of generated motions. For Beat Alignment Score (BAS), CV-CMGM reaches 0.214, a 9.18% improvement over M2S-GAN, and significantly higher than LSTM, CNN-LSTM, and GAN. These results indicate that the Conformer architecture combined with the cross-modal generator accurately captures the correspondence between musical rhythm and conducting motions, yielding motions that are temporally aligned and structurally consistent with the music. Overall, CV-CMGM surpasses all baselines in motion realism, diversity, and music–motion synchronization, confirming its effectiveness and suitability for virtual choir conducting scenarios.

Qualitative analysis based on the visualizations in Fig. 6 further highlights the differences among the models. In the CV-CMGM column, skeletal motions appear smooth and natural, with appropriate arm and hand amplitudes, continuous motion trajectories, and clear synchronization with the musical rhythm, reflecting a highly coherent conducting style consistent with real conductors. By contrast, LSTM-generated motions are rigid, with limited arm swings, poor continuity, fragmented trajectories, and delayed beat responses, indicating that sequential modeling alone cannot capture complex conducting motion dynamics. CNN-LSTM shows improvement in local motion details, but overall motions remain less coherent, hand trajectories are not smooth, and rhythm alignment with music is limited, suggesting that local convolution enhances local precision but fails to capture global motion structure. GAN-generated motions display improved dynamic amplitude but exhibit unnatural jitter and disordered hand trajectories, reflecting insufficient constraints on motion continuity and rhythm. M2S-GAN generates motions that are overall more natural than GAN, with improved beat response and more reasonable arm movements; however, long-sequence continuity and fine motion details remain inferior to CV-CMGM, and minor deviations in hand trajectories persist. CV-CMGM clearly outperforms all other models in motion smoothness, skeletal coherence, beat responsiveness, and motion diversity, consistent with the quantitative results in Table 2; Fig. 5. To evaluate subjective performance, 10 professional conductors rated the generated motions from CV-CMGM, LSTM, CNN-LSTM, GAN, and M2S-GAN based on understandability, dynamic expressiveness, musical structure alignment,

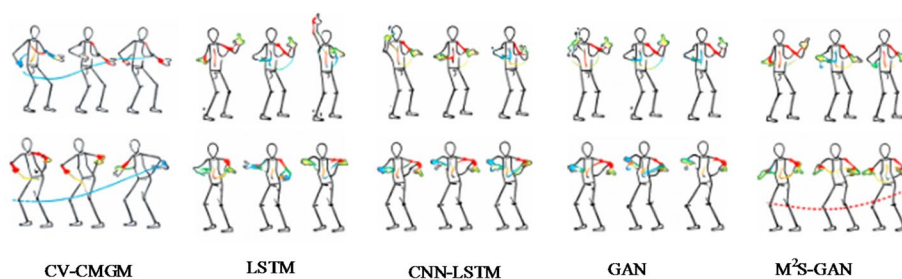
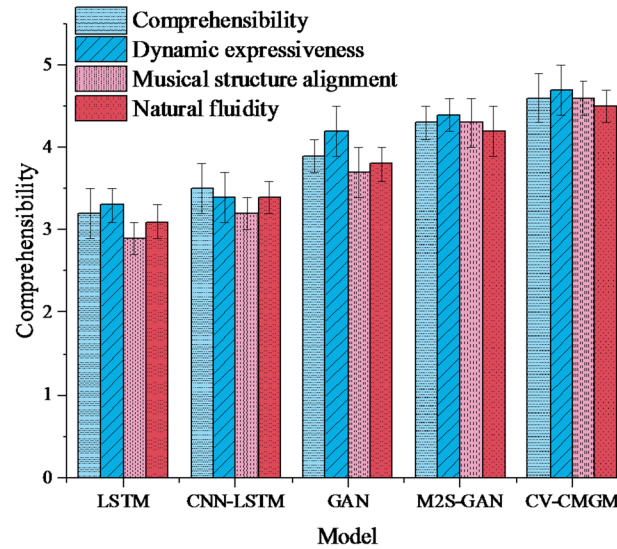


Fig. 6 Qualitative visualization of generated virtual conducting motions

Table 3 User evaluation of generated virtual conducting motions

Model	Understandability	Dynamic expressiveness	Musical structure alignment	Naturalness/fluency
LSTM	3.2 ± 0.3	3.0 ± 0.2	2.9 ± 0.2	3.1 ± 0.2
CNN-LSTM	3.5 ± 0.3	3.3 ± 0.3	3.2 ± 0.3	3.4 ± 0.2
GAN	3.9 ± 0.2	3.8 ± 0.3	3.7 ± 0.3	3.8 ± 0.2
M2S-GAN	4.3 ± 0.2	4.4 ± 0.2	4.3 ± 0.3	4.2 ± 0.3
CV-CMGM	4.6 ± 0.3	4.7 ± 0.3	4.6 ± 0.2	4.5 ± 0.2

**Fig. 7** Comparison of user experiment results

and naturalness/fluency. The mean and standard deviation of the scores are shown in Table 3; Fig. 7.

As seen in Table 3; Fig. 7, the virtual conducting motions generated by CV-CMGM received the highest ratings in terms of comprehensibility, dynamic expressiveness, musical structure alignment, and natural fluency, with scores exceeding 4.4 in all categories. This indicates that the generated conducting motions are the most natural, expressive, and better aligned with the music's rhythm.

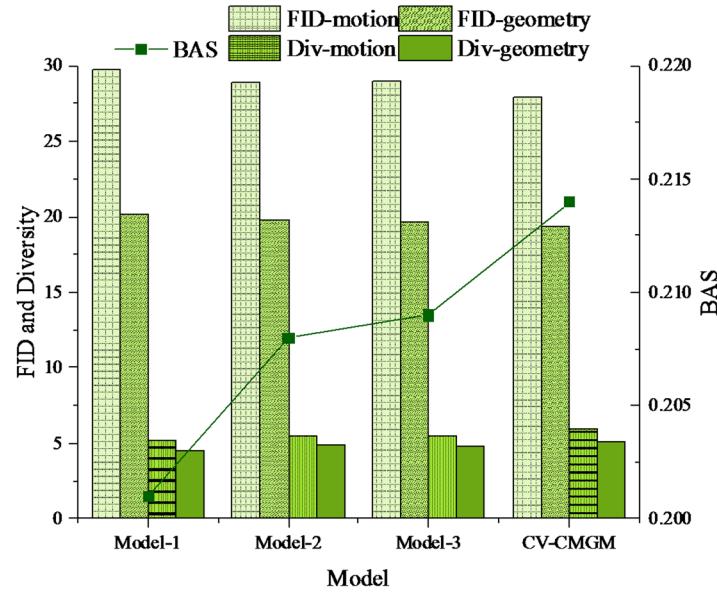
3.2 Ablation study

An ablation study was conducted on the CV-CMGM model, including Model-1, where the Conformer was replaced by a Transformer without convolutional layers; Model-2, where the local temporal discriminator was removed and only the global content discriminator was used; and Model-3, where the global content discriminator was removed and only the local temporal discriminator was used. The evaluation is performed on the ConductorMotion100 dataset, and the results are summarized in Table 4; Fig. 8.

As shown in Table 4; Fig. 8, Model-1 exhibited a slight decline in both FID and diversity, indicating that the convolutional-enhanced Transformer contributes to performance improvement. Compared to CV-CMGM, both Model-2 and Model-3 showed a slight decrease in the quality of the generated conducting motions, suggesting that both the local temporal discriminator and the global content discriminator play crucial roles

Table 4 Ablation study results on conductormotion100

Model	FID-Motion	FID-Geom	Div-Motion	Div-Geom	BAS
Model-1	29.78	20.12	5.23	4.55	0.201
Model-2	28.85	19.78	5.51	4.87	0.208
Model-3	28.96	19.64	5.46	4.82	0.209
CV-CMGM	27.96	19.35	5.94	5.12	0.214

**Fig. 8** Comparison of ablation results of the model structure**Table 5** Ablation study of loss functions on conductormotion100

Model	FID-Motion	FID-Geom	Div-Motion	Div-Geom	BAS
w/o L_{GCN}	29.18	20.43	5.21	4.68	0.205
w/o L_{GAN}	28.67	19.87	5.38	4.79	0.209
w/o L1	28.95	20.05	5.33	4.71	0.207
w/o L_{fm}	28.54	19.95	5.42	4.85	0.211
CV-CMGM	27.96	19.35	5.94	5.12	0.214

in enhancing the quality of the generated motions. Therefore, the synergistic interaction of the components in the CV-CMGM model is vital for improving its performance.

To investigate the contribution of each loss term in the CV-CMGM model for generating high-quality virtual conducting motions, ablation experiments were conducted by removing individual losses. Specifically, four variants were created by excluding the perceptual loss L_{GCN} , GAN loss L_{GAN} , L1 loss, and feature matching loss L_{fm} , respectively. The full model was used as the reference for comparison. Quantitative results are summarized in Table 5, and qualitative visualizations are shown in Fig. 9.

Based on Table 5; Fig. 9, each loss term contributes significantly to the quality of the generated virtual conducting motions. Removing the perceptual loss L_{GCN} leads to lower FID values and reduced motion diversity, indicating its key role in maintaining skeletal structure and preserving natural motion details. Excluding the GAN loss L_{GAN} decreases the overall realism of the motions, resulting in slightly stiff movements and reduced dynamic expressiveness. Omitting the L1 loss causes deviations in motion amplitude

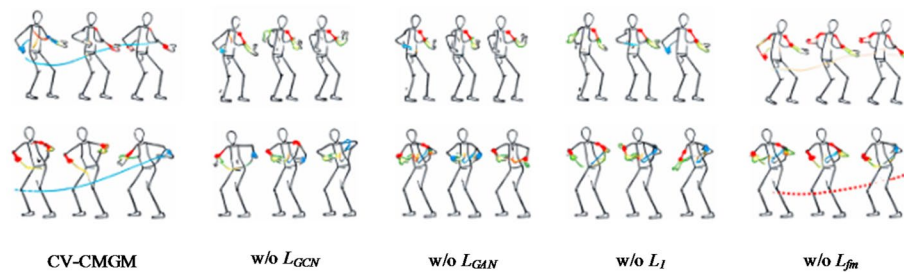


Fig. 9 Visualization of the ablation study on loss functions

relative to the ground truth, slightly lowering motion accuracy. Removing the feature matching loss L_{fm} impairs long-sequence synchronization with the music and reduces beat alignment. Overall, each loss addresses a specific aspect of motion quality. Their combined effect ensures that the generated virtual conducting motions achieve optimal naturalness, structural coherence, dynamic expressiveness, and music alignment. The full CV-CMGM model outperforms all single-loss ablated variants across all evaluation metrics, demonstrating the indispensable role of each loss term in enhancing model performance.

4 Discussion

The proposed CV-CMGM is capable of generating high-quality, natural, and diverse virtual conducting gestures while achieving precise alignment with musical rhythm. Compared with existing studies, CV-CMGM demonstrates clear advantages. Li et al. [26] addressed the problem of “multiple gestures corresponding to the same audio” by proposing a cross-modal latent code splitting strategy. They also designed a relaxed motion loss and a cycle-consistency constraint to optimize variational autoencoder training, aiming to enhance motion diversity. In contrast, CV-CMGM leverages the Conformer architecture to capture both local and global dependencies in music and motion sequences. Combined with a cross-modal generator and dual discriminators, it ensures diverse gesture generation while maintaining precise musical alignment and natural fluidity. Qi et al. [27] proposed the EmotionGesture framework, designed an emotion-beat mining module to extract emotion and audio beat features and model their association, and then used an initial pose based spatiotemporal prompter to generate spatiotemporal coherent pose prompts. Subsequently, the Transformer was used to generate gestures to solve the problem of ignoring emotions when generating voice gestures in virtual avatars. Compared with this approach, CV-CMGM is specifically designed for choral conducting. By integrating gesture generation with real-time assisting applications, it enhances the interpretability and expressiveness of motions in practical music teaching and performance contexts. Roy et al. [28] focused on robotic expressive motion generation and proposed a large language model-based autonomous mapping framework to achieve “intended expression-motion parameter” mapping. CV-CMGM is designed for virtual choral conducting. It generates gestures that are closely aligned with the music while enhancing their naturalness and diversity through local temporal and global content discriminators. This design makes the model particularly well suited for online virtual conducting assistance.

Although the three studies above focus on co-speech gesture or robotic expression generation, their objectives differ from those of choral conducting motion generation.

Co-speech gestures are typically non-deterministic and driven by speech content, whereas choral conducting motions strictly depend on musical beats and melodic structures. Nevertheless, these studies provide valuable technical references for cross-modal motion generation, motion diversity constraints, and modeling of emotional or rhythmic features. This study draws on their approaches to cross-modal generator design and temporal–global dependency modeling, applying them to the generation of virtual choral conducting motions conditioned on music. This enables high-precision alignment between motions and music while maintaining natural expressiveness. Therefore, while this study draws on technical ideas from co-speech gesture generation, its objectives and application are fully focused on choral conducting. The generated motions strictly follow explicit beat constraints. They are also designed to support pedagogical use. Overall, CV-CMGM demonstrates advantages in gesture generation quality, musical alignment, interpretability, and practical application potential. These results highlight the innovation and application prospects of Conformer-based virtual conductor motion generation in intelligent choral conducting.

5 Conclusion

5.1 Summary of research findings

This study introduces the Conformer-based virtual conducting gesture generation model (CV-CMGM) and presents an intelligent online virtual technology-based choral conducting assistance model built upon this framework. The following conclusions are drawn from the experimental results:

- (1) On the ConductorMotion100 dataset, CV-CMGM outperforms the baseline models LSTM, CNN-LSTM, GAN, and M2S-GAN in motion FID, geometric FID, motion diversity, geometric diversity, and BAS. This indicates that the generated motions achieve superior performance in overall motion distribution, spatial structure, action richness, and musical beat responsiveness.
- (2) User experiments reveal that the virtual conducting gestures produced by CV-CMGM received high ratings from professional conductors for comprehensibility, dynamic expressiveness, musical structure alignment, and natural fluency. These findings suggest that the generated gestures exhibit strong practical utility and expressiveness.
- (3) Ablation experiments further show that the Conformer architecture, together with the local temporal and global content discriminators, plays a key role in enhancing motion quality, temporal continuity, and music alignment. Additionally, each loss term contributes to different aspects of performance.

In summary, the CV-CMGM model represents a significant advancement in virtual conducting gesture generation and demonstrates considerable potential for practical applications. However, several limitations remain. Firstly, the scarcity of choral conducting gesture datasets may constrain the model's ability to generalize in certain scenarios. Secondly, while the generated gestures are of high quality, the model's real-time performance and computational efficiency require further refinement. Future research should focus on expanding the dataset, optimizing model efficiency, and incorporating multi-modal information to enhance the model's adaptability and effectiveness in more complex environments.

5.2 Current and future development

With the continuous advancement of virtual technology, artificial intelligence, and multimodal interaction techniques, intelligent virtual conducting systems hold broad prospects in music education, remote rehearsals, and immersive performances. Future development trends include the followings:

- (1) Integrate virtual conductor gesture generation with richer sensory interactions, such as haptic feedback, VR, or augmented reality environments, to enable immersive interactive experiences.
- (2) Explore gesture generation methods for cross-cultural and multi-genre music, allowing the system to adapt to different types of choirs and musical works.
- (3) Leverage cloud computing and distributed inference technologies to support large-scale online choir training and real-time virtual conducting deployment.
- (4) Promote deep integration of virtual conducting technology with educational platforms, music performance software, and intelligent devices, facilitating widespread application in teaching and performance contexts.

Through these advancements, virtual conducting technology is expected to evolve beyond a single-purpose gesture generation tool. It will become an integrated intelligent platform that combines teaching, rehearsal, and performance, providing new innovative approaches for music education and artistic presentation.

Abbreviations

Conformer	Convolution-Enhanced Transformer
GAN	Generative Adversarial Network
VR	Virtual Reality
CNN	Convolutional Neural Network
CV-CMGM	Conformer-based Virtual Conductor Motion Generation Model
MFCC	Mel-Frequency Cepstral Coefficients
HCN	Hierarchical Convolutional Network
Q	Query (in attention mechanism)
K	Key (in attention mechanism)
V	Value (in attention mechanism)
GCN	Graph Convolutional Network
AIST++ Dataset	A Large-Scale Dance Motion Dataset with Music for Deep Learning
LSTM	Long Short-Term Memory
M ² S-GAN	Music Motion Synchronized Generative Adversarial Network
FID	Fréchet Inception Distance
Div	Diversity
BAS	Beat Align Score

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Author contributions

Wenwen Zhou contributed to writing—original draft, writing—review and editing, data curation, formal analysis, project administration, resources, software.

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Data availability

All data generated or analysed during this study are included in this article.

Declarations

Ethics approval and consent to participate

Ethical approval has been obtained for all protocols from the Kyungnam University. Ethics committee to confirm that the study meets national and international guidelines for research on humans. For all studies involving human participants, informed written consent to take part in the research has been obtained prior to the commencement of the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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